Noah Neuweg

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AI & Data Engineer · Applied Math and Data Science, UC San Diego

I build LLM automation and data pipelines end to end, from scraping and cleaning the data to shipping the workflow that runs on it. At Scripps Institution of Oceanography I replaced a multi-service PDF extraction stack with a single OpenAI API workflow in n8n; on my own projects I have trained classifiers on 337,000+ records and run production Python scrapers against live financial data.

SKILLS

Languages	Python, SQL, JavaScript/TypeScript, Bash
AI / LLM	OpenAI API, RAG (chunking, embeddings, FAISS retrieval), prompt engineering, agent workflows, Promptfoo (LLM evals), n8n
ML & Stats	scikit-learn, pandas, NumPy, matplotlib; clustering, EM and maximum-likelihood estimation, classification, regression, feature engineering, cross-validation
Linux	Ubuntu/WSL2, shell scripting, SSH, systemd services, Podman/Docker containers
Automation	n8n, Selenium, BeautifulSoup, scheduled jobs, API integration
Infrastructure	Git, FastAPI, Next.js, Vercel, Cloudflare Workers
Databases	PostgreSQL, MySQL, SQLite (joins, aggregation, schema design), Prisma

EXPERIENCE

AI & Workflow Engineer *Jun 2026 – Sep 2026*

CCHDO, Scripps Institution of Oceanography, UC San Diego — San Diego, CA

- Built Python and n8n automation workflows that extract structured metadata from oceanographic cruise report PDFs, replacing manual data entry for the CCHDO archive.
- Redesigned the extraction architecture to send PDFs directly to the OpenAI API by URL, eliminating an AWS Textract dependency and a FastAPI intermediary service.
- Deployed and debugged services on a shared UCSD Linux server, working through Podman container networking and FastAPI service configuration.
- Prototyped LLM agent and RAG workflows for document QA and metadata validation; proposed and iterated on automation ideas with the data manager.

Student Computing Support Technician *Dec 2023 – Jun 2026*

Jacobs School of Engineering, UC San Diego — San Diego, CA

- Imaged and deployed 30+ systems per week, including OS installs, driver setup, and software configuration.
- Performed secure multi-pass (3-pass) data sanitization on hundreds of machines before reassignment.
- Diagnosed and repaired hardware and software issues across laptops, desktops, and peripherals in a production support environment.
- Maintained inventory and service logs supporting auditing and lifecycle tracking.

PROJECTS

Treasury Auction Scraper & Rate Tracker *2024 – Present*

- Python scraper for Treasury auction results using Selenium for JavaScript-rendered pages and BeautifulSoup for HTML parsing.
- Parsed and standardized auction tables into structured CSV datasets; generated time-series plots comparing rate trends across maturities.
- Designed for scheduled runs and threshold-based alerting.

League of Legends Match Outcome Predictor *2023 – 2025*

- Processed 337,000+ professional match records with 160+ attributes across multiple seasons.
- Built an end-to-end ML workflow in Python (preprocessing, feature engineering, train/test split, evaluation); Random Forest classifier reached 87% test accuracy.
- Identified leakage risk and next steps: time-based splits, backtesting, and probability calibration.

n8n Content Automation Pipeline *2024 – Present*

- Automated content pipeline in n8n integrating API calls, scheduled workflows, and file handling to produce short-form video outputs.
- Modular ingestion, transformation, and publishing steps with logging and failure handling for repeatable runs.

EDUCATION

B.S. Mathematics — Applied Data Science, University of California, San Diego *Expected Dec 2026*

Relevant coursework: Data Management (DSC 100), Data Visualization (DSC 106), Probabilistic ML (DSC 140A), Algorithms & Graph Theory (DSC 40A/B), Applied Data Science Lifecycle (DSC 80), Applied Linear Algebra (MATH 102), Real Analysis (MATH 142A/B), Stochastic Processes (MATH 180C)

Andrew Ng Machine Learning Specialization, Coursera *Feb 2026*

ACTIVITIES

Scuba certified (Open Water) | Gardening | Running and hiking | Sailing (Hobie, Quests/Sabots)